



Thermal phase noise in giant interferometric fiber optic gyroscopes

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Abstract: The thermal phase noise in giant interferometric fiber optic gyroscopes (fiber length $L > 10$ km) and its impact on the detection sensitivity are theoretically derived and experimentally verified. It is confirmed that thermal phase noise cannot be overlooked for the giant IFOGs. Utilizing high order eigen frequency modulation can effectively suppress the walk-off component of thermal phase noise, but the residual part contributes to high-frequency range thus limits the detection bandwidth of giant IFOGs. The self-noise is experimentally demonstrated as $3.5 \text{ nrad/s}/\sqrt{\text{Hz}}$ at low frequencies and $5.2 \text{ nrad/s}/\sqrt{\text{Hz}}$ at 100 Hz in the IFOG with a 30-km single mode fiber coil. Discussions about the fiber characteristics on thermal phase noise are presented, which paves the way to the design of giant IFOGs.

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1. Introduction

The interferometric fiber optic gyroscope (IFOG) is an inertial sensor for detecting rotational velocity with high sensitivity as well as stability, and hence, has been widely utilized in many civil and military applications [1–4]. The excellent performance of IFOGs comes from the long and large fiber coils that sense and accumulate small Sagnac phase shift into detectable signals, and the conventional navigation grade IFOGs usually have fiber coil in length of 1–2 km.

It could be interesting and practically useful to ask if the fiber coil can be even longer to achieve better performance. For this purpose, a series of studies have been made. A reference grade IFOG was demonstrated using fiber coil with the enclosed area of $\sim 1000 \text{ m}^2$, reaching a sensitivity of $4.6 \text{ nrad/s}/\sqrt{\text{Hz}}$ [3]. Clivati et al. [5] utilized a fiber communication network to get an enclosed area of 20 km^2 with the sensitivity of $10 \text{ nrad/s}/\sqrt{\text{Hz}}$. A series of giant IFOGs [6–8] are designed to detect weak rotational seismic motions. Moreover, a road map of giant IFOG is proposed [9], in which fiber coil length of 15 km and diameter of 4 m is planned to reach a sensitivity of $\sim 0.3 \text{ nrad/s}/\sqrt{\text{Hz}}$.

We specifically define the “giant IFOGs” are those IFOGs with fiber coil length more than 10 km. The application scenarios of giant IFOGs fall into two categories. One is for high-end navigation and guidance purpose, in which the integration of rotation matters a lot, and hence, the bias stability performance is more concerned [10]. The other one is for ultra-small rotation detection, such as rotational seismic motion recording and building monitoring [8, 11–13]. In these applications, the sensitivity of IFOGs in bandwidth range from 0.01 to 100 Hz is more concerned [13].

Although the sensitivity of giant IFOGs is leveraged by the large enclosed area, it is not necessarily true that the signal-to-noise ratio (SNR) in rotation detection also improves since the noise increases with the fiber length as well. The common knowledge of the noises in IFOGs needs to be revisited by considering the giant coil. The polarization nonreciprocity (PN) error degrades the long-term stability and is mainly determined by the polarization extinction ratio (PER) of the polarizer as well as the coherence length of light source, but not the length of fiber coil [14, 15]. However, since the cross coupling between orthogonal polarizations

increases with the fiber length, the PN error would slightly increase. On the other hand, the excess relative intensity noise (RIN) is the major cause of short-term noises but it doesn't depend on the fiber length. Besides, the shot noise, originating from the occurrence independence of the photons, is inversely proportional to the square root of the detected light power [16–18]. The light attenuation in long fiber will lower the light intensity at output thus increase the shot noise. There should be a trade-off between enclosed area and detected power, which gives a sensitivity limit [19, 20]. The Shupe effect and non-reciprocal noise introduced by thermal stress increase with the fiber length [21, 22], but it could be controlled by maintaining the environment temperature. Apparently, the electronic noise has no relationship with the optical structure.

The thermal phase noise, introduced by random thermal fluctuations in the waveguide, degrades the performance of fiber optic interferometers. Such noise significantly elevates the noise floor of fiber optic Mach-Zehnder interferometers, overwhelms the performance limit of shot or quantum noise [23–25]. A series of investigation have also been carried out on the thermal phase noise in optical fiber Sagnac interferometers [26–29]. In low and medium grade gyroscopes, the thermal phase noise is lower than RIN even shot noise [30], therefore, it was not regarded as the dominant noise in most IFOGs. However, since the thermal phase noise increases with the length of the waveguide (fiber), it brings more severe impact to the giant IFOGs that need to be carefully addressed.

In this work, we theoretically analyzed and experimentally verified the characteristics of thermal phase noise in giant IFOGs. The thermal phase noise appears in the vicinity of every odd harmonic of modulation frequency. The noise tailing (we refer it as to “walk-off” terms) from neighborhood frequencies significantly elevates the noise floor, but it can be well suppressed by using high order eigen frequency modulation. The residual thermal phase noise degrades the sensitivity of giant IFOG in high frequency range. Since the thermal phase noise increases with the fiber length, the SNR would not monotonously improve by lengthening the coil. The relationship between thermal phase noise and fiber characteristics is also discussed for giant IFOGs design.

2. Thermal phase noise in IFOGs

The thermal phase noise is a kind of fundamental noise in optical fiber, introduced by the inevitable thermodynamic fluctuations of the waveguide when the temperature is above 0 K. The thermal phase noise origins from the fluctuation of the refractive index of the fiber due to the atomic thermal motion. Glenn pointed out that the thermal phase noise is very similar to the Johnson–Nyquist noise in electrical circuits, and can be derived from the optical Nyquist theorem [31].

The output of an IFOG can be represented as

$$I_d(t) = \frac{1}{2} I_0(t) \{1 + \cos[\phi_S(t) + \phi_N(t) + \phi_m(t)]\} \quad (1)$$

where $I_0(t)$ and $I_d(t)$ are the light intensities at input and detector, respectively; $\phi_S(t)$, $\phi_N(t)$ and $\phi_m(t)$ are the Sagnac phase shift, thermal phase noise, and modulation phase, respectively. For IFOGs using sinusoidal modulation, $\phi_m(t) = \phi_{m0} \cos(\omega_m t)$, where ϕ_{m0} is the modulation depth and ω_m is the modulation frequency. The power spectral density (PSD) of light intensity owing

to thermal phase noise is derived as follows:

$$\text{PSD}_{I_{dN}}(\omega) = \pi I_0^2 S(\omega), \quad (2a)$$

$$S(\omega) = \sum_{n=0}^{\infty} J_{2n+1}^2(\phi_{m0}) \{ \langle \Delta\phi_{N,\text{rms}}^2[\omega + (2n+1)\omega_m] \rangle + \langle \Delta\phi_{N,\text{rms}}^2[\omega - (2n+1)\omega_m] \rangle \}, \quad (2b)$$

$$\langle \Delta\phi_{N,\text{rms}}^2(\omega) \rangle = \frac{k_B T^2 L_c}{\kappa \lambda^2} \left(\frac{dn_{\text{eff}}}{dT} + n_{\text{eff}} \alpha_L \right)^2 \times \ln \left[\frac{\left(\frac{2}{W_0} \right)^4 + \left(\frac{\omega}{D} \right)^2}{\left(\frac{4.81}{d} \right)^4 + \left(\frac{\omega}{D} \right)^2} \right] \times \left[1 - \text{sinc} \left(\frac{\omega L_c n_{\text{eff}}}{c} \right) \right] \quad (2c)$$

In Eq. (2b), J_n is the n th order Bessel function. Equation (2c) depicts the spectral density of phase noise in detection introduced by the thermal phase noise in fiber-optic Sagnac interferometers [23, 26, 28]. The thermal phase noises near D.C. are modulated to every odd harmonics of ω_m and weighted with Bessel functions (detailed derivation can be found in [26]). Here k_B denotes the Boltzmann constant. dn_{eff}/dT is the temperature coefficient of the effective refractive index of fiber. The fiber is characterized as mode field radius of W_0 , cladding diameter of d , total length of L_c , linear thermal expansion coefficient of α_L , with thermal conductivity κ and thermal diffusivity D . λ is the optical wavelength and c is the light speed in vacuum. The physics interpretation of Eqs. (2) is presented in Appendix I.

In harmonics demodulation, only the noise near modulation frequency matters. The noise terms in Eq. (2b) can be divided into two categories:

(1). The $J_1^2(\phi_{m0}) \langle \Delta\phi_{N,\text{rms}}^2(\omega - \omega_m) \rangle$ term accompanies with the modulation frequency f_m . Derived from Eq. (2c) that, it becomes quite small when closed to f_m , and hence, would not contribute long-term noise. We denote this noise as *modulation-frequency-term*. As further discussed in detail in Section 4, this term would raise the noise of IFOG in high-frequency range.

(2). The rest terms except that of Term (1) (namely $\omega - \omega_m$ term) are centered at the odd harmonics of f_m but their falling tails walk off to the neighborhood harmonics and also increase the noise near f_m . We denote it as *walk-off-terms* and will be discussed in Section 3.

It's necessary to clarify the detection bandwidth requirement of IFOGs in different application scenarios. For the usage in which bias stability matters more, such as inertial navigation and guidance [3, 10], the bandwidth is commonly within 10 Hz. For some other applications, such as rotational seismology, exploration, and building monitoring, the bandwidth needs up to 100 Hz. Then, in the discussions involved in this work, the low-frequency noise is defined as noise in D.C. to 10 Hz; while the high-frequency is between 10 Hz to 100 Hz.

The minimum detectable rotation rate (MDRR) limited by thermal phase noise is derived as

$$\sqrt{\text{PSD}_{\Omega_N}(\omega)} = \frac{\lambda c}{4\pi L_c R_c} \cdot \sqrt{2\pi} \cdot \frac{\sqrt{S(\omega + \omega_m)}}{J_1(\phi_{m0})} \quad (3)$$

where R_c is the average radius of fiber coil.

According to Eqs. (2), the influence of thermal phase noise increases with the fiber length. However, the MDRR limited by other kinds of noise, such as shot noise, RIN and electronic noise of the circuit, is $\propto L_c^{-1}$. Therefore, although the thermal phase noise is not severe for short coil IFOGs (see calculations below for details), such noise will become one of the most dominant noise in giant IFOGs.

3. Walk-off noise and its suppression

As discussed before, the thermal phase noise cannot be overlooked in giant IFOGs. The schematic of giant IFOG is illustrated in Fig. 1 for experiment investigation of thermal phase noise. In

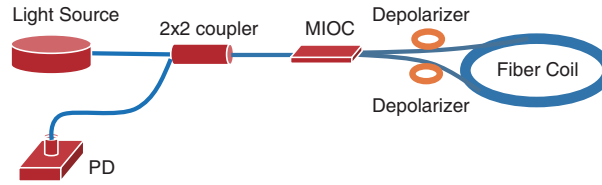


Fig. 1. Schematic diagram of the giant IFOG adopted in this paper.

this setup, a 30630-m long, single mode (SM) fiber coil is adopted and the IFOG operates in depolarized “minimum configuration” scheme.

More specifically, a low degree of depolarization (DOP), broadband amplified spontaneous emission (ASE) light source with central wavelength of 1550 nm and bandwidth of 40 nm is adopted, of which power is 200 mW. Two depolarizers with DOP less than 0.2% are put on both arms of fiber coil to avoid signal fading. The output light intensity is detected by the photodiode (PD) and acquired with sample rate of 10 million samples per second (MSPS). The modulation with eigen (or proper) frequency $f_e = 1/2\tau_g$ or its odd harmonics should be applied to the IFOG to reduce the Rayleigh backscattering error [32,33], in which τ_g is the group transit time of the fiber coil. For the fiber coil adopted in this work, the eigen frequency is calculated as $f_e = 3.3$ kHz. The modulation depth is set to 2.7 for RIN suppression [34,35].

The spectrum at the PD output combined with simulation noises is presented in Fig. 2(a). The overall noise is composed of thermal phase noise, RIN, shot noise and electronic noise, among which, the thermal phase noise is calculated by Eq. (3) with parameters listed in Table 2. The other noises are estimated according to [18], which are frequency-independent in the concerned frequency range. It's worthy to point that the insert loss of the fiber coil is 6.5 dB (0.21 dB/km), and the shot noise of this setup is calculated as 0.18 nrad/s/ $\sqrt{\text{Hz}}$, which is much smaller than the measured RIN (3.5 nrad/s/ $\sqrt{\text{Hz}}$). Therefore, the shot noise still has limited contribution to the overall noise in our setup even a giant fiber coil is adopted. The measurement and simulation agree well with each other. Apparently, the measured spectrum is not flat, implies the thermal phase noise is considerably large in the giant IFOG.

A detailed noise spectrum in vicinity of modulation frequency $f_m = f_e$ is shown in Fig. 2(b), in which the comparison of two distinct terms of thermal phase noise, i.e. modulation-frequency-term and walk-off-terms, as well as the contributions from other noises (RIN, shot, electronic) are also illustrated. Since the required detection bandwidth is about several hundreds of Hertz, the walk-off-terms contribute dominant noise in the concerned bandwidth, degrade both angular random walk (ARW) and BI performance significantly. In other words, the thermal phase noise walk-off-terms are the leading noise in the giant IFOG, and need to be suppressed to enhance the sensitivity.

We proposed that the walk-off-terms can be effectively suppressed by high-order eigen frequency modulation. As derived from Eq. (2c) and shown in Fig. 2, each term in Eq. (2b) has severe noise tail falling from its center frequency, and the noise decreases when leaving away from the center. By adopting higher order of eigen frequency modulation, the interval between modulation harmonics is much larger, and hence, less noise walks off toward the neighborhood harmonics that reduce the thermal phase noises. Noticed that we should choose the odd harmonics of eigen frequency for the modulation. From a physics point of view, it is equivalent to the case that, the thermal phase noise is remarkably reduced when the thermal diffusion length δ at the modulation frequency smaller than the mode diameter. (see Appendix I and [36,37] for details.)

Figure 2(c) shows the noise spectrum in the vicinity of modulation frequency at 33rd of eigen frequency ($f_m = 33f_e = 110$ kHz), in which the walk-off-terms are remarkably lower than that in Fig. 2(b). Since the walk-off-terms have been effectively suppressed, the RIN becomes the new

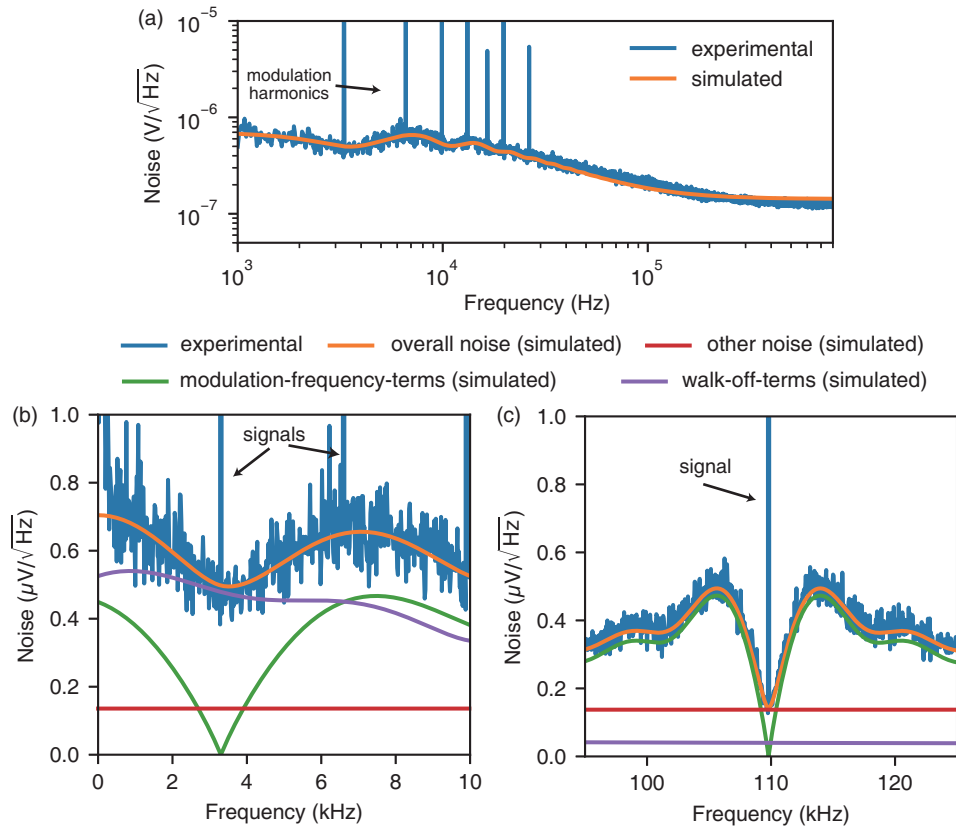


Fig. 2. Experimental and simulated noise spectrum: (a) and (b). $f_m = f_e$ (3.3 kHz); (c). $f_m = 33f_e$ (110 kHz).

leading noise. It implies that, in this case, the noise performance of giant IFOG in low-frequency range is the same with that of regular-size IFOGs since no extra low-frequency noise is involved.

Figure 3 presented the dependence of self-noise at 1 Hz on the modulation frequency. As discussed, the modulation-frequency-term is negligible in contributing low-frequency noise. Thus the self-noise at 1 Hz mainly comes from the walk-off-terms and RIN. The ARW due to RIN is independent of modulation frequency while the walk-off-terms decreases with modulation frequency. As shown in Fig. 3, the simulation matches well with the experiment, validated our theoretical model. It is noteworthy that, for our giant IFOG, modulation frequency above 100 kHz (31st or higher order of f_e) is regarded as sufficiency to overcome the walk-off-terms.

For a detailed comparison, the static tests with the modulation frequency of 3.3 kHz ($f_m = f_e$) and 110 kHz ($f_m = 33f_e$) are performed and the results are shown in Fig. 4(a). The Allan deviation (ADEV) curves and root PSD of the two tests are illustrated in Figs. 4(b) and (c), accordingly. With walk-off-terms suppression, the ARW decreases to $1.2 \times 10^{-5} \text{ }^\circ/\sqrt{\text{h}}$ and BI reaches $4.2 \times 10^{-5} \text{ }^\circ/\text{h}$, implying the effectiveness of high order eigen frequency modulation in walk-off-terms suppression. Notice that the bump in 2 ~ 6 Hz in Fig. 4(c) comes from environmental variation, because the same frequency of noise is also detected by an accelerometer put besides the IFOG.

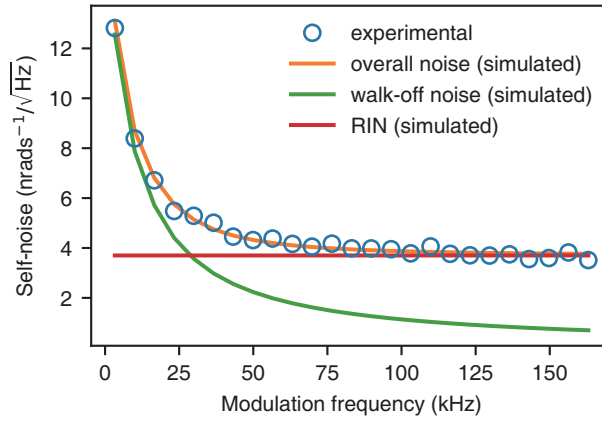


Fig. 3. The relationship between self-noise at 1 Hz and modulation frequency: experiment and simulation results.

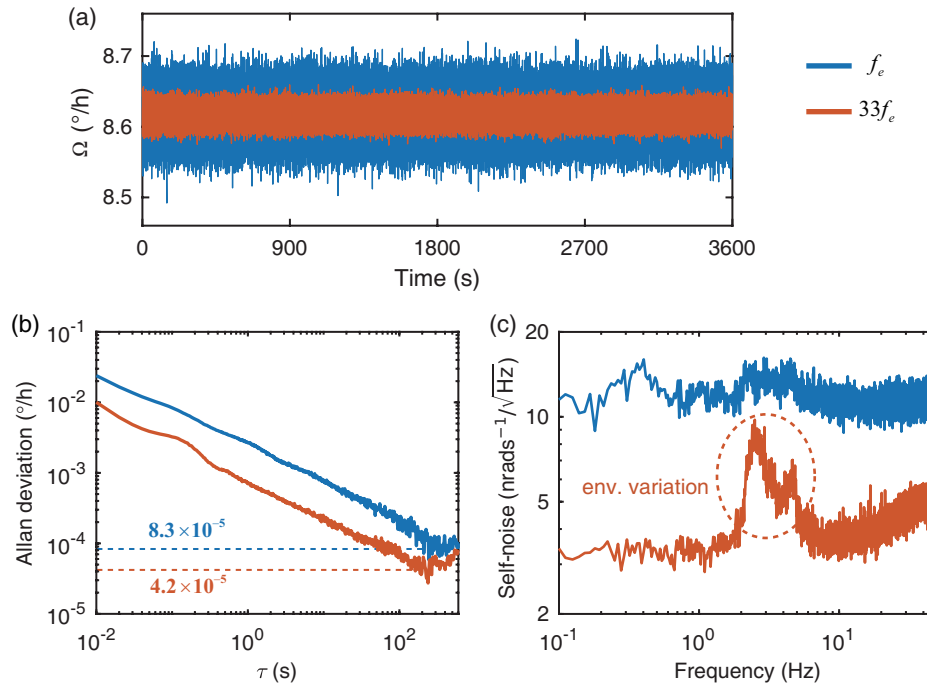


Fig. 4. (a) The giant IFOG outputs with the modulation frequency of f_e and $33f_e$; the (b) ADEV curves and (c) root PSD of the results in (a).

4. Residual thermal phase noise in high-frequency range

By utilizing high order eigen frequency modulation, the walk-off-terms of thermal phase noise can be suppressed even lower than RIN. Therefore, the residual thermal phase noise is contributed by the modulation-frequency-term. As discussed above, the residual modulation-frequency-term doesn't harm the long-term stability performance but will introduce noises into the high-frequency range. In this case, the residual thermal phase noise limited MDRR can be rewritten from Eq. (3)

Table 1. Comparison of ARW, BI, and Self-noise.

Modulation frequency	ARW ($^{\circ}/\sqrt{\text{h}}$)	BI ($^{\circ}/\text{h}$)	self-noise ($\text{rad/s}/\sqrt{\text{Hz}}$)
f_e (3.3 kHz)	4.4×10^{-5}	8.3×10^{-5}	1.2×10^{-8}
$33f_e$ (110 kHz)	1.2×10^{-5}	4.2×10^{-5}	3.5×10^{-9}

as:

$$\sqrt{\text{PSD}_{\Omega_N}(\omega)} = \frac{\lambda c}{4\pi L_c R_c} \cdot \sqrt{2\pi \langle \Delta\phi_{N,\text{rms}}^2(\omega) \rangle} \quad (4)$$

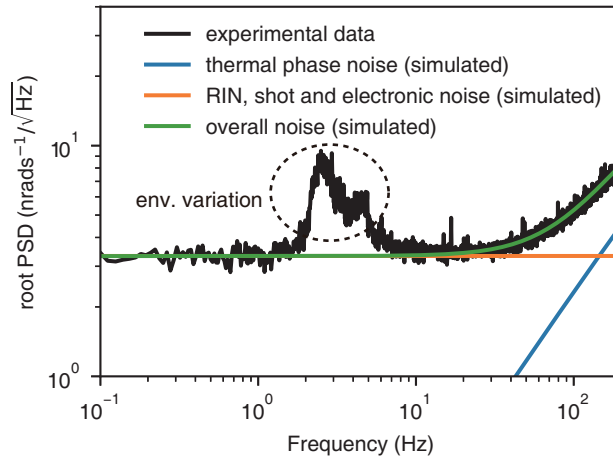


Fig. 5. The root PSD of the IFOG's outputs ($f_m = 33f_e$).

Figure 5 shows the experiment and simulation results of self-noise when 33rd eigen frequency modulation is applied. The simulation agrees well with the experiment except the bump of environmental variation noise. The noise spectrum is almost flat when the frequency is lower than 10 Hz while the noise increases with frequency between 10 ~ 200 Hz. The residual thermal phase noise elevates the overall noise significantly above ~ 10 Hz and becomes dominant portion above ~ 100 Hz, which will certainly lower the sensitivity and limit the detection bandwidth.

5. Discussions on thermal phase noise suppression and design of giant IFOGs

As discussed above, the walk-off-terms are the leading noise in giant IFOGs with harmonic modulation. For square wave modulation, such noises also exist but have different amplitudes [29]. Even so, the walk-off-terms are avoidable by applying the high order eigen frequency modulation. However, the modulation-frequency-term is inevitable since it is indistinguishable from the Sagnac phase shift.

The thermal phase noise originates from the atomic thermal motion and is proportional to the temperature. The most straightforward way to suppress thermal phase noise is to lower the temperature. For the giant IFOG in this work, if the temperature was lower than several tens of Kelvins, the thermal phase noise will not degrade the sensitivity below 100 Hz. Most commercial fiber products and optical components are guaranteed to work as low as -60°C , even lower operating temperature is possible by special design. The IFOGs deployed at fixed stations can adopt cryo-temperature control to suppress the thermal phase noise, while it is not suitable for portable sensors.

The material properties in Eq. (2c), such as thermal expansion coefficient, thermal diffusivity and thermal conductivity are determined by SiO_2 molecule itself and can not be adjusted easily [24,38]. They keep constant values in different types (single mode, polarization maintaining, etc.) of fibers, regardless of the difference in cladding or mode field diameter. The temperature coefficient of refractive index can hardly change as well even refractive index profile and dopant is applied [30,39]. However, as shown in Eq. (2c), thermal phase noise can be reduced by engineering the fiber with mode diameter and cladding diameter. Figure 6 presents the thermal phase noise level (at 100 Hz) with different types of fibers (Corning and Yangtze Optical Fibre and Cable Company, YOFC) but keeping the rest parameters consistent with the experiment setup. From the perspective of thermal phase noise suppression, fiber with larger mode diameter and smaller cladding diameter is preferred in the design of giant IFOGs.

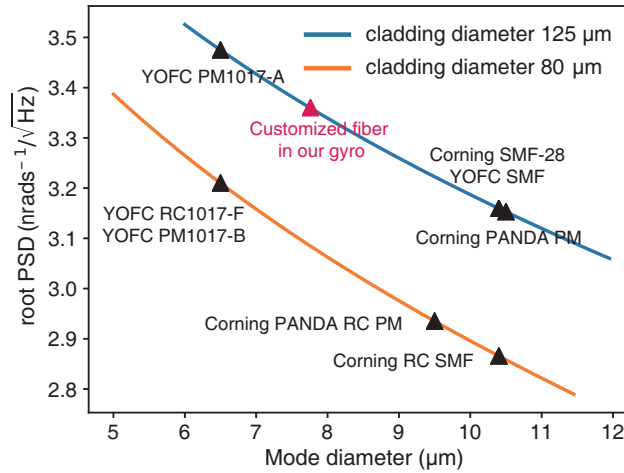


Fig. 6. self-noise (at 100 Hz) of the IFOG owing to thermal phase noise for fiber with different mode and cladding diameters: a simulation.

The dependence of phase noise on fiber length is illustrated in Fig. 7(a). The lines (orange and blue) indicates the phase noise of commercial IFOGs (iXblue) with and without RIN suppression [9]. Such noise level is consistent with Honeywell [3] and Optolink [40]'s products, which represent the state-of-art noise level of commercial products. Self-noise at different frequencies is calculated and shown in Fig. 7(b).

As shown in Fig. 7, the thermal phase noise increases with the fiber length. For the IFOGs with the parameters listed in Table 2, the noise at 100 Hz has a minimum value of 3.1 nrad/s/√Hz with fiber length of ~17 km. The minimum noise varies with fiber parameters but the trends are similar. The SNR at 100 Hz is limited by the residual thermal phase noise and cannot be further improved by simply lengthening the fiber coil. Then, for the applications that operate at 100 Hz or higher frequency, such as exploration seismology, engineering seismology or planetary seismology [13], the fiber coil length needs to be optimized to achieve the best SNR.

The thermal phase noise at 10 Hz also increases but is still smaller than other noise for a fiber coil shorter than 50 km. Therefore, for the applications that only long-term stability matters or operate below 10 Hz, the thermal phase noise has limited impact.

6. Conclusions

In this work, thermal phase noise in giant IFOGs is theoretically investigated and experimentally verified. The thermal phase noise origins from the inevitable thermodynamic fluctuations of the fiber and increases with the fiber length. The theoretical model shows that such noise can be

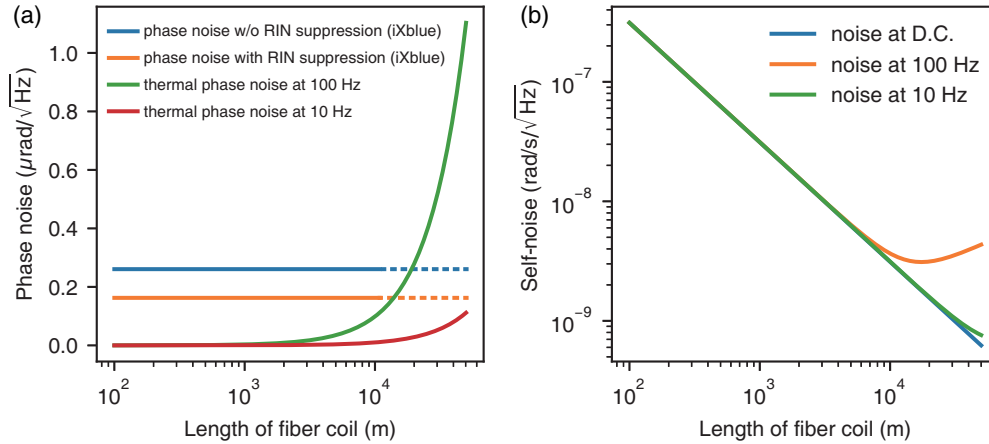


Fig. 7. (a) Phase noise and (b) overall self-noise for fiber coils with different length. In (b), the adopted residual RIN, shot and electronic noise values are from iXblue's products with RIN suppression [9].

divided into two categories, i.e., modulation-frequency-term and walk-off-terms, respectively. The walk-off-terms of thermal phase noise contribute remarkably large noise and degrade the sensitivity and stability performance of giant IFOGs, however, it can be effectively suppressed by adopting the high order eigen frequency modulation. The residual thermal phase noise, i.e., modulation-frequency-term, induces noise to high-frequency range of giant IFOGs and limits the detection bandwidth.

We adopted a giant IFOG prototype with a fiber coil of 30 km to verify the theoretical model. The self-noise is detected as $3.5 \text{ nrad/s}/\sqrt{\text{Hz}}$ at low frequencies and $5.2 \text{ nrad/s}/\sqrt{\text{Hz}}$ at 100 Hz. The relationship between fiber characteristics and thermal phase noise is discussed, and we suggest that the fibers with larger mode field diameter and smaller cladding diameter will benefit the thermal phase noise suppression. For the giant IFOG operating at 100 Hz, the noise level reaches the minimal value at an optimized fiber length but not monotonously drops with lengthening the fiber length. For the IFOGs operating within 10 Hz, the thermal phase noise is not the major noise and can be omitted in the IFOG design.

Appendix I

Here we present an interpretation of Eq. (2c) from a physics point of view. Equation (2c) depicts the spectrum of non-reciprocal thermal phase noise in Sagnac interferometers, which is composed of 4 terms:

- Term 1: $k_B T^2 / (\kappa \lambda^2)$ is the basic term from optical Nyquist effect [31] with the Boltzmann constant.
- Term 2: $dn_{\text{eff}}/dT + n_{\text{eff}} \alpha_L$ is the thermo-optic coefficient, describing the change of effective refractive index with response to temperature.
- Term 3: $\ln \left\{ \left[\left(\frac{2}{w_0} \right)^4 + \left(\frac{\omega}{D} \right)^2 \right] / \left[\left(\frac{4.81}{d} \right)^4 + \left(\frac{\omega}{D} \right)^2 \right] \right\}$ is a dimensionless function of frequency [23] where the constant 4.81 comes from the insulation boundary condition. This term gives low-frequency noise while approaches to 0 at high frequency. The spectrum of this term could be found in Fig. 1 of [23].

- Term 4: $L_c [1 - \text{sinc}(\omega L_c n_{\text{eff}}/c)]$ is the term from the Sagnac interferometer [27] and behaves as a delay-line filter

As elaborated by Logozinskii [36], Term 3 is actually an exponential decaying harmonic thermal waves, which can be depicted by a thermal diffusion length $\delta = \sqrt{2D/\omega}$. In silica, $D \approx 1 \text{ mm}^2/\text{s}$, which yields $\delta(1 \text{ kHz}) \approx 5 \text{ }\mu\text{m}$ and $\delta(10 \text{ kHz}) \approx 2 \text{ }\mu\text{m}$. As explained by Duan [37], thermal fluctuations become uncorrelated above the depth δ , and when it becomes smaller than the optical mode diameter (about $10 \text{ }\mu\text{m}$), these fluctuations start to be averaged out over the section of the mode which reduces the actual thermal phase noise.

Appendix II

The parameters of the fiber coil applied in the simulations are listed in Table 2, in which the optical, thermal and structural parameters are included.

Table 2. Values of Parameters of the Experimental IFOG Applied in the Simulation.

Parameters	Property	Value
T	Temperature ²	287 K
n_{eff}	Effective refractive index ¹	1.467
α_L	Thermal expansion coefficient [38]	$5.5 \times 10^{-7} / \text{K}$
D	Thermal diffusivity	$0.82 \times 10^{-6} \text{ m}^2/\text{s}$
κ	Thermal conductivity	$1.37 \text{ W} / (\text{m} \cdot \text{K})$
λ	Wavelength ¹	1550 nm
dn_{eff}/dT	Temperature coefficient of refractive index ¹	$1.1 \times 10^{-5} / \text{K}$
$2W_0$	Mode field diameter ¹	$7.7 \text{ }\mu\text{m}$
d	Cladding diameter of the fiber ¹	$125 \text{ }\mu\text{m}$
L_c	Coil length ¹	30,630 m
R_c	Average coil radius ¹	0.19 m
P_o	Detected light power ²	$310 \text{ }\mu\text{W}$
η_D	Photoelectric conversion efficiency of the PD ¹	0.90

¹ from manufacturer's data-sheet; ² measured in the setup.

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